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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

# 3 - First Order Logic - Solutions

3.1) Let $\mathcal{L}$ be a first order logic language, such that $\mathcal{F}_0 = \{\text{ed}, \text{patsy}, \text{carl}, \text{monique}\}$ and $\mathcal{R}_2 = \{F, M, H, S, B\}$ Let $F(x, y)$ mean that $x$ is the father of $y$; $M(x, y)$ denotes $x$ is the mother of $y$. Similarly, $H(x, y)$, $S(x, y)$ and $B(x, y)$ will represent the relations husband/sister/brother of, respectively. Translate the following sentences into first order logic:

(a) Everybody has a mother.

(b) Everybody has a father and a mother.

(c) Whoever has a mother has a father.

(d) Ed is a grandfather.

(e) Nobody's grandmother is anybody's father.

(f) Ed and Patsy are husband and wife.

(g) Carl is Monique's brother-in-law.

**Solution:**

(a) $\forall x \exists y\ M(y, x)$

(b) $\forall x \exists y \exists z\ (M(y, x) \land F(z, x))$

(c) $\forall x (\exists y\ M(y, x) \rightarrow \exists z\ F(z, x))$

(d) $\exists x \exists y\ (F(\text{ed}, x) \land (F(x, y) \lor M(x, y)))$

(e) $\forall x (\exists y \exists z\ (M(x, y) \land (F(y, z) \lor M(y, z))) \rightarrow \forall y \neg F(x, y))$

(f) $H(\text{ed}, \text{patsy})$

(g) $\exists x ((B(\text{carl}, x) \land H(x, \text{monique})) \lor (H(\text{carl}, x) \land S(x, \text{monique})))$

3.2) Translate each of the following propositions by a sentence in Portuguese or English; consider the set $\mathbb{R}$ of real numbers as the universe.

(a) $\exists x \forall y\ (x + y = y)$

(b) $\forall x \forall y\ (((x \geq 0) \land (y < 0)) \rightarrow (x - y > 0))$

(c) $\exists x \exists y\ (((x \leq 0) \land (y \leq 0)) \land (x - y > 0))$

**Solution:**

(a) There exists a neutral element for addition on real numbers.

(b) The difference between a non-negative real number and a negative real number is positive.

(c) There exist non-positive real numbers whose difference is positive.

3.3) Which of the following strings are formulas in first order logic? Specify a reason for failure for strings that are not.

a) Let $\mathcal{F}_0 = \{\mathfrak{m}\}$, $\mathcal{F}_1 = \{\mathfrak{f}\}$ and $\mathcal{R}_2 = \{\mathsf{S}, \mathsf{B}\}$:

(a) $\mathsf{S}(\mathfrak{m}, \mathfrak{x})$

(b) $\mathsf{B}(\mathfrak{m}, \mathfrak{f}(\mathfrak{m}))$

(c) $\mathfrak{f}(\mathfrak{m})$

(d) $\mathsf{B}(\mathsf{B}(\mathfrak{m}, \mathfrak{x}), \mathfrak{y})$

(e) $\mathsf{S}(\mathsf{B}(\mathfrak{m}), z)$

(f) $(\mathsf{B}(\mathfrak{x}, \mathfrak{y}) \to (\exists z\, \mathsf{S}(z, \mathfrak{y})))$

(g) $(\mathsf{S}(\mathfrak{x}, \mathfrak{y}) \to \mathsf{S}(\mathfrak{y}, \mathfrak{f}(\mathfrak{f}(\mathfrak{x}))))$

(h) $(\mathsf{B}(\mathfrak{x}) \to \mathsf{B}(\mathsf{B}(\mathfrak{x})))$

**Solution:**

| | **Formula?** | |
|---|---|---|
| (a) | ✓ | |
| (b) | ✓ | |
| (c) | × | $\mathfrak{f}(\mathfrak{m})$ is a term, not a formula |
| (d) | × | an atomic formula $\mathsf{B}(\mathfrak{m}, \mathfrak{x})$ cannot occur inside another formula |
| (e) | × | atomic formula $\mathsf{B}(\mathfrak{m})$ should have two parameters and not occur inside another formula |
| (f) | ✓ | |
| (g) | ✓ | |
| (h) | × | atomic formula $\mathsf{B}(\mathfrak{x})$ should have two parameters and not occur inside another formula |

b) Let $\mathcal{F}_0 = \{\mathfrak{c}, \mathfrak{d}\}$, $\mathcal{F}_1 = \{\mathfrak{f}\}$, $\mathcal{F}_2 = \{\mathfrak{g}\}$, $\mathcal{F}_3 = \{\mathfrak{h}\}$ and $\mathcal{R}_3 = \{\mathsf{P}, \mathsf{Q}\}$:

(a) $\forall \mathfrak{x}\, \mathsf{P}(\mathfrak{f}(\mathfrak{d}), \mathfrak{h}(\mathfrak{g}(\mathfrak{c}, \mathfrak{x}), \mathfrak{d}, \mathfrak{y}))$

(b) $\forall \mathfrak{x}\, \mathsf{P}(\mathfrak{f}(\mathfrak{d}), \mathfrak{h}(\mathsf{P}(\mathfrak{x}, \mathfrak{y}), \mathfrak{d}, \mathfrak{y}))$

(c) $\forall \mathfrak{x}\, \mathsf{Q}(\mathfrak{g}(\mathfrak{h}(\mathfrak{x}, \mathfrak{f}(\mathfrak{d}), \mathfrak{x}), \mathfrak{g}(\mathfrak{x}, \mathfrak{x})), \mathfrak{h}(\mathfrak{x}, \mathfrak{x}, \mathfrak{x}), \mathfrak{c})$

(d) $\exists z\, (\mathsf{Q}(z, z, z) \to \mathsf{P}(z))$

(e) $\forall \mathfrak{x} \forall \mathfrak{y}\, (\mathfrak{g}(\mathfrak{x}, \mathfrak{y}) \to \mathsf{P}(\mathfrak{x}, \mathfrak{y}, \mathfrak{x}))$

(f) $\mathsf{Q}(\mathfrak{c}, \mathfrak{d}, \mathfrak{c})$

**Solution:**

| | **Formula?** | |
|---|---|---|
| (a) | × | relational symbol $\mathsf{P}$ should have three parameters |
| (b) | × | relational symbol $\mathsf{P}$ should have three parameters and not occur inside a term |
| (c) | ✓ | |
| (d) | × | relational symbol $\mathsf{P}$ should have three parameters |
| (e) | × | $\mathsf{g}(\mathsf{x},\mathsf{y})$ is not a formula |
| (f) | ✓ | |

3.4) Let $\mathcal{L}$ be a first order logic language such that $\mathcal{F}_1 = \{\mathsf{f}\}$, $\mathcal{F}_2 = \{\mathsf{g}\}$ and $\mathcal{R}_2 = \{\mathsf{P},\mathsf{Q}\}$. Let $\varphi$ be $\exists \mathsf{x}\,(\mathsf{Q}(\mathsf{x},\mathsf{y}) \wedge (\forall \mathsf{y} \neg \mathsf{Q}(\mathsf{y},\mathsf{x}) \vee \exists \mathsf{z}, \mathsf{P}(\mathsf{y},\mathsf{z}))))$.

a) Identify the free and bound occurrences of variables in $\varphi$.

**Solution:** The occurrences of $\mathsf{x}$ are bound and the occurrence of $\mathsf{z}$ is also bound. Whereas the first occurrence of $\mathsf{y}$ is free, the second is bound, and the third is free again.

b) Is there a variable in $\varphi$ that has both free and bound occurrences?

**Solution:** Variable $\mathsf{y}$ occurs both free and bound in the formula.

c) Consider the terms $w$, $\mathsf{f}(\mathsf{x})$ and $\mathsf{g}(\mathsf{y},\mathsf{z})$:

(a) Compute $\varphi[\mathsf{x} \mapsto w]$, $\varphi[\mathsf{y} \mapsto w]$, $\varphi[\mathsf{y} \mapsto \mathsf{f}(\mathsf{x})]$ and $\varphi[\mathsf{y} \mapsto \mathsf{g}(\mathsf{y},\mathsf{z})]$.
**Solution:**

$$\begin{aligned}
\varphi[\mathsf{x} \mapsto w] &= \varphi \\
\varphi[\mathsf{y} \mapsto w] &= \exists \mathsf{x}\,(\mathsf{Q}(\mathsf{x},w) \wedge (\forall \mathsf{y} \neg \mathsf{Q}(\mathsf{y},\mathsf{x}) \vee \exists \mathsf{z}, \mathsf{P}(w,\mathsf{z})))) \\
\varphi[\mathsf{y} \mapsto \mathsf{f}(\mathsf{x})] &= \exists \mathsf{x}\,(\mathsf{Q}(\mathsf{x},\mathsf{f}(\mathsf{x})) \wedge (\forall \mathsf{y} \neg \mathsf{Q}(\mathsf{y},\mathsf{x}) \vee \exists \mathsf{z}, \mathsf{P}(\mathsf{f}(\mathsf{x}),\mathsf{z})))) \\
\varphi[\mathsf{y} \mapsto \mathsf{g}(\mathsf{y},\mathsf{z})] &= \exists \mathsf{x}\,(\mathsf{Q}(\mathsf{x},\mathsf{g}(\mathsf{y},\mathsf{z})) \wedge (\forall \mathsf{y} \neg \mathsf{Q}(\mathsf{y},\mathsf{x}) \vee \exists \mathsf{z}, \mathsf{P}(\mathsf{g}(\mathsf{y},\mathsf{z}),\mathsf{z}))))
\end{aligned}$$

(b) Which of $w$, $\mathsf{f}(\mathsf{x})$ and $\mathsf{g}(\mathsf{y},\mathsf{z})$ are free for $\mathsf{x}$ in $\varphi$?
**Solution:** All of $w$, $\mathsf{f}(\mathsf{x})$ and $\mathsf{g}(\mathsf{y},\mathsf{z})$ are free for $\mathsf{x}$ in $\varphi$. Since $\mathsf{x}$ is bound in $\varphi$, we know that any substitution for $\mathsf{x}$ will keep the formula the same, so no new ound variables will occur.

(c) Which of $w$, $\mathsf{f}(\mathsf{x})$ and $\mathsf{g}(\mathsf{y},\mathsf{z})$ are free for $\mathsf{y}$ in $\varphi$?
**Solution:** The term $w$ is free for $\mathsf{y}$. Both the terms $\mathsf{f}(\mathsf{x})$ and $\mathsf{g}(\mathsf{y},\mathsf{z})$ are not, since the free variable in $\mathsf{f}(\mathsf{x})$ becomes bound in $\varphi$, and the same happend to the variable $\mathsf{z}$ in $\mathsf{g}(\mathsf{y},\mathsf{z})$.